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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

November 2022 - Paper 2 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-FS-Maths-L2-Paper-2-November-2022

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Do not write outside the box

Section A

Answer **all** questions in the spaces provided.

1 Circle the integer.

[1 mark]

0.5 $\frac{1}{8}$ 7 −10.2

Q1 Worked Solution:

3 =

[Method & Accuracy]

2 Write 9 507 211 in words.

[1 mark]

Answer _____

Q2 Worked Solution:

3 = _____

[Method & Accuracy]

_____

3 Work out 3 years to 9 months as a ratio.

Give your answer in its simplest form.

[2 marks]

Q3 Worked Solution:

8 = _____

[Method & Accuracy]

_____

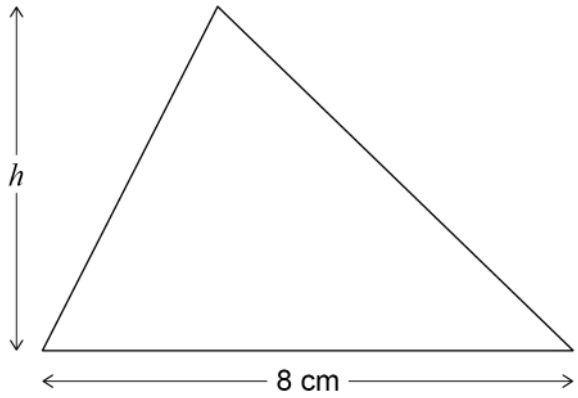
Answer _____ : _____

Do not write outside the box

7 A triangle has an area of 20 cm^2

The base of the triangle is 8 cm

Not drawn accurately



Work out the perpendicular height, h, of the triangle.

[2 marks]

Answer _____ cm

8 Calculate $2(7 + 3k)$ when $k = -1.8$

[2 marks]

Q8 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Answer _____

12

Turn over ►

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

9 Lorry driving

Asha is a lorry driver.

9 (a) Asha drives from Southampton to Leeds.

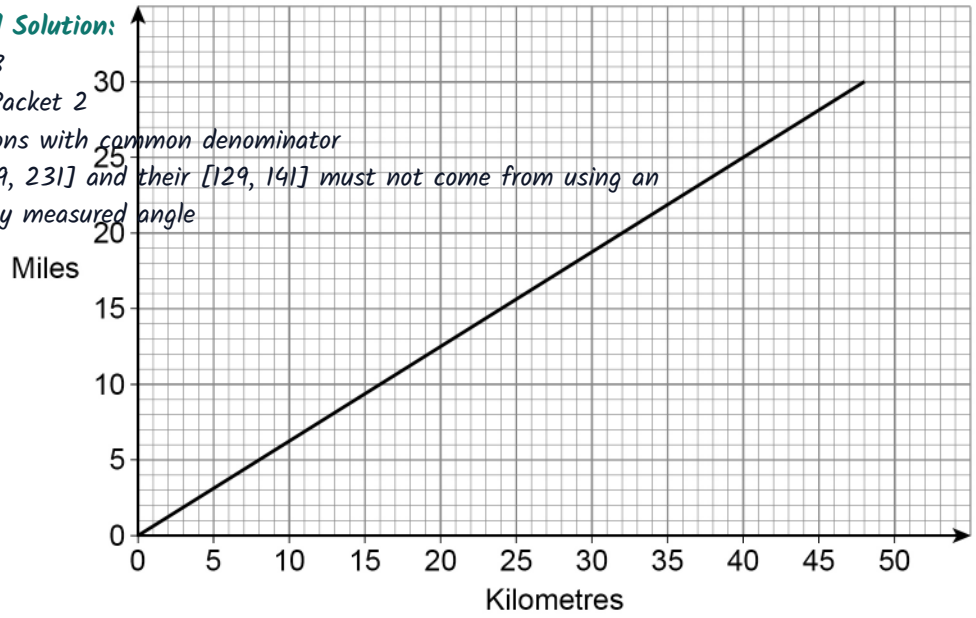
The journey

is 380 kilometres

takes 5 hours and 30 minutes.

The graph can be used to convert between miles and kilometres.

Q9 Worked Solution:
24 and 8
24 and Packet 2
oe fractions with common denominator
their [219, 231] and their [129, 141] must not come from using an incorrectly measured angle
[A1]



Asha works out that his average speed is **over** 40 mph

Is he correct?

You **must** show your working.

[4 marks]

Do not write outside the box

9 (b) The amount Asha is paid each week is calculated using the formula

$$P = 0.73(0.14d + 65n)$$

where

P = pay in pounds

d = distance driven in kilometres that week

n = number of days worked that week

Last week Asha worked for 5 days.

His pay for last week's work was £605.17

How many kilometres did Asha drive last week?

[4 marks]

Answer _____ kilometres

8

Turn over for the next question

Do not write outside the box

10 Fundraising

Carol is fundraising for a sports club.

10 (a) Carol designs a game.

The game uses

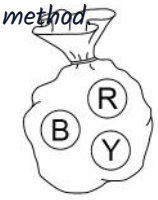
a bag containing a red ball, a blue ball and a yellow ball

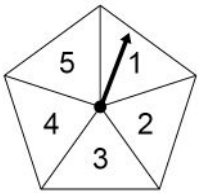
a fair, 5-sided spinner.

Q10 Worked Solution:

See official AQA mark scheme method

[Method & Accuracy]





The player

picks a ball at random from the bag

and

spins the spinner.

The player wins if they pick the **red** ball and the spinner lands on an **even** number.

Carol says,

"The chance of winning is **more than** 10%"

Is she correct?

You **must** show your working.

[4 marks]

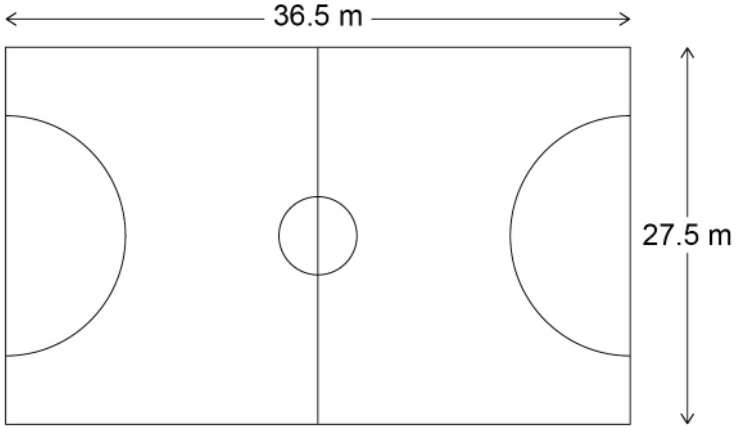
Do not write outside the box

10 (b) The club wants to use some of the money to paint the lines on a mini football pitch.

The lines to be painted are

- the four sides of a rectangle measuring 27.5 m by 36.5 m
- a halfway line measuring 27.5 m
- a centre circle with a radius of 1.5 m
- two semicircles, each with a radius of 8 m

Not drawn accurately



It costs £3.25 per metre to paint the lines.

In total, how much will it cost to paint all the lines?

[6 marks]

Answer £ _____

Turn over ►

Do not write outside the box

10 (c) After painting the lines the club has £8225

They invest $\frac{2}{7}$ of this money in a bank account for 4 years.

The account pays compound interest at 3% per year.

Is the investment worth **more than** £2700 at the end of the **4** years?

You **must** show your working.

[4 marks]

14

Do not write outside the box

11 **Ice cream**

Suzi has an ice cream van.

11 (a) Suzi buys tubs of ice cream and sells scoops of ice cream.

Each scoop is in the shape of a sphere with radius 2.8 cm

$$\text{volume of sphere} = \frac{4}{3}\pi r^3$$

r = radius of sphere

Suzi buys 5-litre tubs.

1 litre = 1000 cm^3

Suzi wants to buy enough tubs to sell at least 200 scoops.

Work out how many tubs Suzi should buy.

You **must** show your working.

[5 marks]

Q11 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Answer _____

Question 11 continues on the next page

Do not write outside the box

11 (b) Suzi uses a 15% discount voucher when she buys the tubs of ice cream.

She pays £76.50 after the discount.

Suzi says,

"I save **less than** £14 by using the discount voucher."

Show working to support this statement.

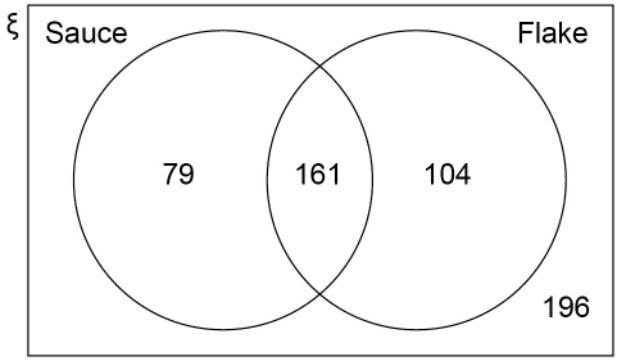
[3 marks]

Do not write outside the box

11 (c) Suzi sells two ice cream toppings, sauce and flake.

She hopes that the probability that a customer, picked at random, buys **at least one** topping will be more than $\frac{7}{10}$

The Venn diagram shows what toppings the customers buy over one weekend.



Over this weekend, does Suzi achieve the probability she hopes to get?

You **must** show your working.

[3 marks]

11

Turn over for the next question

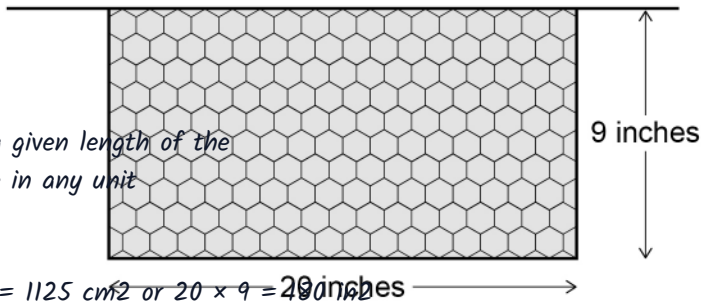
Do not write outside the box

12 **Bees**

Mary keeps bees and sells the honey they produce.

12 (a) The bees live in a beehive.

Mary's beehive holds rectangular frames full of honeycomb.



Not drawn accurately

Q12 Worked Solution:
their 12 × 8
their 12 cannot be a given length of the honeycomb or frame in any unit
96
Using overall areas
20 × 9 × 2.5 × 2.5 = 1125 cm2 or 20 × 9 = 180 in2
and dividing by total area of the honeycomb piece

Each frame measures 9 inches by 20 inches.

The beehive holds 8 frames.

Mary cuts the honeycomb into rectangular pieces measuring 11 cm by 7.5 cm

Work out the maximum number of pieces that Mary can get from her **beehive**.

Use 1 inch = 2.5 cm

[5 marks]

11 × 7.5 = 82.5 cm2 or (11 ÷ 2.5) × (7.5 ÷ 2.5) = 13.2in2
giving answers of 13.63... or rounded to 13
could continue to score the final method mark eg 13.63 × 8 = 109.04
[A1] [M1] [A1]

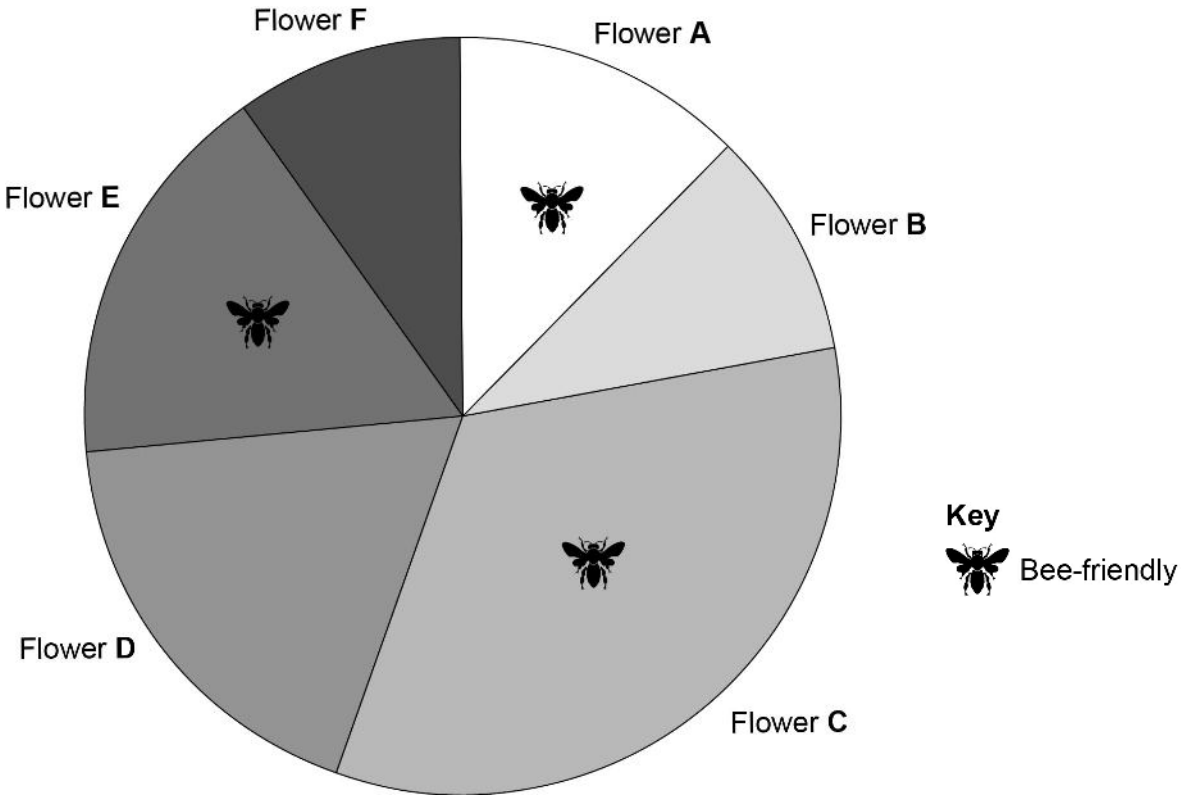
Answer _____

Do not write outside the box

12 (b) Mary wants to grow some bee-friendly flowers.

She finds information about the different flowers produced from two packets of seeds.

Packet 1



Two thirds of the seeds in **Packet 2** produce bee-friendly flowers.

Mary wants to buy the packet producing the greater proportion of bee-friendly flowers.

Which packet should she buy?

You **must** show your working.

[4 marks]

Do not write outside the box

12 (c) Here are the instructions for planting flower seeds.

Use 4 grams of seed per square metre of garden.

Mix the seeds with sand in the ratio

mass of seed : mass of sand = 2 : 5

Mary measures her neighbours' gardens to work out the average-sized garden.

Garden (m^2)	Frequency	Midpoint	
$0 < \text{area} \leqslant 10$	2		
$10 < \text{area} \leqslant 20$	8		
$20 < \text{area} \leqslant 30$	12		
$30 < \text{area} \leqslant 40$	3		

Estimate the total mass of seed and sand mix needed to cover an average-sized garden.

[6 marks]

Answer _____ grams

15

END OF QUESTIONS

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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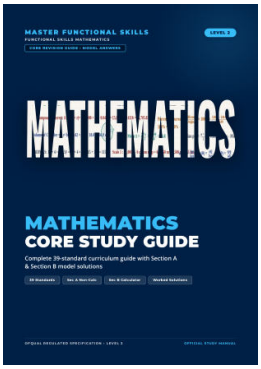
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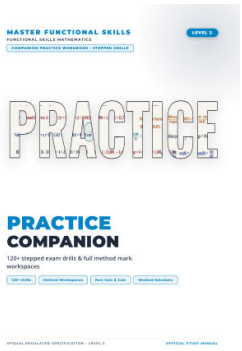


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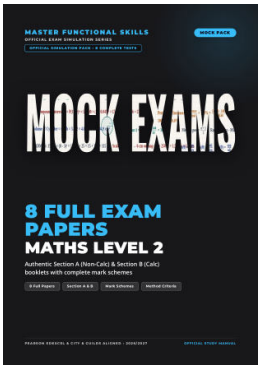


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